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Surviving the Fintech Disruption

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1 Big picture

- **Question.** What happens to labor demand, occupational structure, firm innovation, and firm performance when fintech technologies disrupt existing finance-related tasks?
- **Core setting.** The paper studies the recent fintech wave, which combines finance and information technology and often arrives from outside traditional finance through payments, blockchain, cybersecurity, robo-advising, peer-to-peer lending, mobile transactions, and data analytics.
- **Measurement innovation.** The authors build a new occupation-level fintech exposure measure by comparing the text of occupational tasks in O*NET to the text of fintech patent titles and abstracts.
- **Main labor result.** Occupations with higher fintech exposure experience declines in job postings and employment. The baseline effect is economically large: moving from the 20th to the 80th percentile of exposure implies a meaningful loss of job postings in an average occupation-year cell.
- **Not just general technology.** Fintech exposure has weak correlation with AI, software, offshorability, routine-task, and abstract-task exposure. It is a distinct technology shock rather than a simple relabeling of automation or AI.
- **Creative-destruction result.** Fintech destroys some jobs but creates others. New suboccupations are more likely to appear in fintech-exposed areas, especially blockchain, cybersecurity, and peer-to-peer technologies.
- **Complementarity versus substitution.** The aggregate negative effect hides heterogeneity. Occupations whose tasks are complementary to fintech can experience growing demand even inside sectors where other exposed jobs contract.
- **Upskilling result.** Fintech exposure raises demand for hybrid “finance + technology” skills, technology skills, work experience, and bachelor’s-degree requirements. The disruption therefore changes not only job quantities but also skill composition.
- **Firm adjustment result.** Firms partly absorb the shock by changing hiring, reallocating workers internally, increasing separations from obsolete roles, and redirecting innovation toward fintech.
- **Firm performance result.** Firms do better when they have either internal fintech innovation capability or workforces whose tasks are complementary to fintech. Innovative firms can gain employment, sales, and productivity.
- **Economic takeaway.** Fintech is harmful for many individual workers in exposed occupations but less harmful, and sometimes beneficial, for adaptive firms. The firm can reconfigure tasks and innovation, while workers bear more of the reallocation cost.

2 Data and measurement

2.1 Fintech patent data

- The fintech innovation corpus comes from fintech patents. Patent titles and abstracts are treated as descriptions of the technological content of innovations.
- The fintech patent data cover multiple technological categories, including cybersecurity, mobile transactions, data analytics, blockchain, peer-to-peer technology, robo-advising, and Internet-of-Things-related fintech.
- Fintech exposure is time-varying because patent activity changes across years and categories. The main measure uses a five-year rolling window of fintech patents.

Interpretation: Patents are used as an ex ante technology-shock measure. This is important because the paper wants to know which occupations were exposed before observing job-market outcomes.

2.2 Occupation task data

- The authors use O*NET task descriptions to characterize the task content of each occupation.
- The occupation-level measure aggregates task-level scores using O*NET task weights based on importance, relevance, and frequency.
- The baseline occupation classification follows the occupation coding used in Autor-Dorn/Webb-style labor-market work, which helps compare fintech exposure with general technology exposure measures.

Interpretation: The paper measures exposure at the task level rather than the industry level. This allows fintech to affect an occupation even outside the finance sector if its tasks overlap with fintech patent language.

2.3 Burning Glass job postings

- The main labor-demand data are online job postings from Burning Glass Technologies for 2010–2018.
- The occupation-year sample contains 2,288 occupation-year cohorts and more than 100 million vacancies.
- The authors also use firm-occupation-year postings to study within-firm adjustments and sectoral heterogeneity.
- Job postings are used to measure both the number of jobs and the changing skill requirements inside jobs.

Interpretation: Job postings are useful because they reveal firms’ demand for labor and skills before employment fully adjusts. This is especially helpful for technology shocks that change hiring needs quickly.

2.4 Employment and wage validation

- The paper validates job-posting results using Occupational Employment and Wage Statistics employment and wage data.
- Employment is the equilibrium outcome after labor demand, labor supply, and frictions interact.
- Wages are used to determine whether exposure changes compensation in addition to job quantity.

Interpretation: Employment effects are smaller than job-posting effects, which is consistent with the idea that job postings respond faster and more directly than realized employment.

2.5 Firm-level and worker-flow data

- The paper links job postings to firms and industries to study how firms respond to fintech exposure.
- LinkedIn employment histories are used to study internal transfers and separations across occupations.
- Patent data are used again at the firm level to measure fintech innovation through invented and acquired fintech patents.
- Compustat firm outcomes are used to measure employment, sales, and total factor productivity.

Interpretation: This structure lets the paper distinguish a worker-level effect from a firm-level effect. Individual occupations can shrink while adaptive firms grow.

3 Models and empirical specifications

3.1 Constructing fintech exposure

Let $FT_{o,t}$ denote fintech exposure of occupation o in year t . The paper aggregates task-level exposure scores as

$$FT_{o,t} = \sum_{i \in o} \omega_i FT_{o,i,t} = \sum_{i \in o} \omega_i \sum_{j \in A_t} FT_{o,i,j,t}, \quad (1)$$

where ω_i is the weight of task i inside occupation o , and A_t is the set of fintech patent applications in the relevant rolling window.

Interpretation: An occupation is more exposed when its important tasks are textually close to a large amount of recent fintech patenting.

3.2 Baseline occupation-year labor demand

The main occupation-level specification is

$$\Delta Y_{o,t} = \beta_1 FT_{o,t-1} + \beta_2 X_o + \gamma_t + \varepsilon_{o,t}, \quad (2)$$

where $\Delta Y_{o,t}$ is the year-over-year log change in job postings, positions, employment, or wages.

- $FT_{o,t-1}$ is lagged fintech exposure.
- X_o includes competing occupational technology exposure controls: AI, software, routine-task, abstract-task, and offshorability exposure.
- γ_t are year fixed effects.
- Regressions are weighted by underlying job postings or employment.

Interpretation: The coefficient β_1 asks whether occupations exposed to fintech shrink or grow relative to other occupations in the same year, after controlling for other technology exposures.

3.3 Long-difference specification

The paper also estimates long differences over 2010–2014, 2015–2018, and the full period 2010–2018:

$$\Delta^{LD} Y_o = \beta FT_o + \Gamma X_o + u_o. \quad (3)$$

Interpretation: Long differences reduce noise from year-by-year variation and ask whether occupations with higher fintech exposure experience larger cumulative changes.

3.4 Complementarity and substitution

The paper defines whether fintech exposure is complementary or substitutive to an occupation’s existing tasks. A typical interaction specification is

$$\Delta Y_{o,t} = \beta FT_{o,t-1} + \theta (FT_{o,t-1} \times Complementarity_o) + \Gamma X_o + \gamma_t + u_{o,t}. \quad (4)$$

Interpretation: The baseline fintech effect can be negative, but a positive interaction means the decline is weaker or even reversed for occupations whose tasks are helped by fintech rather than replaced by it.

3.5 Suboccupation creation

To measure occupational creation, the paper estimates Probit models where the dependent variable equals one if a suboccupation becomes newly categorized in a given wave:

$$Pr(NewSubocc_{s,t} = 1) = \Phi(\beta FT_{s,t-1} + \Gamma X_s + \lambda_t). \quad (5)$$

Interpretation: This specification captures the creative side of creative destruction: fintech can eliminate old postings while creating new specialized job titles.

3.6 Upskilling specification

For skill requirements, the paper uses Probit models of the form

$$Pr(SkillIncrease_{o,t} = 1) = \Phi(\beta FT_{o,t-1} + \Gamma X_o + \gamma_t), \quad (6)$$

where the outcome indicates whether postings in an occupation increasingly require finance-plus-tech skills, finance skills, technology skills, experience, or education.

Interpretation: The important question is not only whether firms hire fewer workers, but whether the remaining or new jobs require different human capital.

3.7 Firm-level adjustment and performance

Firm-occupation specifications take the generic form

$$\Delta Y_{i,o,t} = \beta FT_{o,t-1} + \Theta' Interactions_{i,o,t} + \alpha_{i,t} + \eta_o + \varepsilon_{i,o,t}. \quad (7)$$

For performance, the paper estimates firm-year regressions such as

$$\Delta Y_{i,t} = \beta FT_{j,t-1} + \theta_1 (FT_{j,t-1} \times Innovator_i) + \theta_2 (FT_{j,t-1} \times ComplementaryWorkforce_i) + Controls + FE + \varepsilon_{i,t}. \quad (8)$$

Interpretation: The firm-level analysis asks whether firms merely reduce labor demand or actively change their internal organization and innovation strategies to survive the disruption.

4 Identification and empirical design

4.1 Source of variation

- The key variation is occupation-level exposure to fintech patents interacted with time-varying patent activity.
- The design compares more exposed occupations to less exposed occupations while controlling for alternative exposure to AI, software, routine work, abstract tasks, and offshoring.
- Year fixed effects absorb aggregate labor-market conditions and economy-wide fintech waves.

Interpretation: The empirical strategy is not a randomized experiment. It is a carefully controlled exposure design that uses pre-existing task content and external fintech patenting to measure differential exposure.

4.2 Why the measure is credible

- The measure is built from task text and patent text before looking at job outcomes.
- The paper shows fintech exposure differs from existing measures of AI, software, routine-task, abstract-task, and offshorability exposure.
- The measure captures both intensity of innovation and occupational relevance.

Interpretation: If fintech exposure were just another proxy for general technology exposure, the paper's contribution would be weaker. The low correlation with other technology exposures is therefore central.

4.3 Cross-validation strategy

- Job-posting results are validated with positions, employment, and wages.
- The paper studies both occupation-year and firm-occupation-year panels.
- It looks at both quantity effects and skill requirements.
- It examines firm innovation and firm outcomes to understand adaptation.

Interpretation: The paper's credibility comes from showing a coherent pattern across many outcomes: postings decline, employment declines more slowly, skill requirements rise, firm internal reallocation increases, and innovative firms perform better.

4.4 What remains hard to prove

- Patent text may not capture every fintech innovation, especially business-model innovation that is not patented.
- Job postings measure demand but not actual hiring perfectly.
- Occupations can change internally, so the same occupation code may hide task redefinition.
- Firm adaptation can confound the direct labor-displacement effect because firms can change both technology and workforce composition.
- Wages may respond slowly or be measured too coarsely to detect occupational effects.

Interpretation: The paper provides strong evidence of technology exposure and adjustment, but the exact welfare effect for workers requires more information on displaced workers' earnings and career paths.

5 Tables and figures

5.1 Figure 1: Measuring fintech exposure

Read:

- The figure shows the construction of the occupation-level fintech exposure measure.
- The upper branch starts from fintech patents, extracts titles and abstracts, and converts them into fintech keywords and scores.
- The lower branch starts from occupations and task descriptions.
- Task scores are aggregated into occupation scores using task weights.

Economic message: The paper’s central variable is not an industry dummy or a subjective classification. It is built from a two-sided text comparison between what fintech patents do and what workers actually do in their jobs.

Interpretation: This matters because fintech can affect nonfinancial occupations and can affect different jobs inside the same firm differently. The task-level measure gives the paper much more granularity than a sector-level shock.

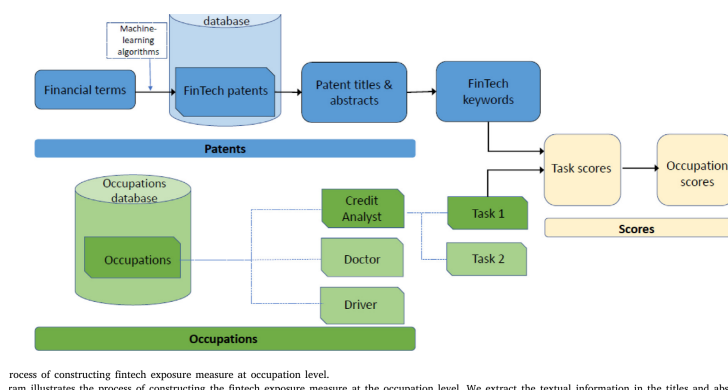


Figure 1: process of constructing fintech exposure.

5.2 Figures 2–3: Fintech patent content and exposure over time

Read:

- Figure 2 shows that early fintech patents emphasize transaction, account, card, payment, information, and data-related language.
- By 2017, mobile, token, server, network, authentication, and device-related terms become more prominent.
- Figure 3 shows that aggregate fintech exposure rises strongly over time.
- Panel B of Figure 3 shows staggered waves: cybersecurity and mobile transactions are early; blockchain and peer-to-peer become more visible later.

Economic message: Fintech is not a one-time technology. It is a sequence of related but distinct shocks. This helps explain why the paper finds stronger effects in later years and across different subsectors.

Interpretation: The word clouds show that the technology is economically close to payment infrastructure, financial authentication, transaction processing, and data handling. Those are exactly the tasks where traditional finance jobs are likely to be disrupted.



Fig. 2. Most frequent keywords in fintech patents
This figure presents the clouds of top keywords in the fintech patents filed in 2003–2006 (Panel A), 2017 (Panel B), and 2003–2017 (Panel C). Font size indicates the frequency of the keywords. We only display the most frequent 75 keywords in each panel.

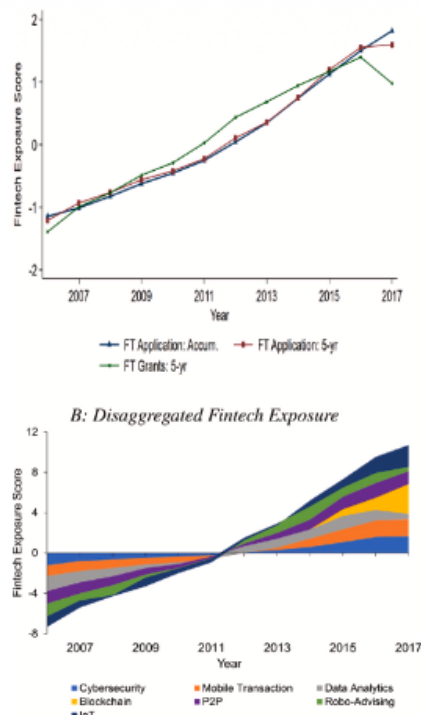


Fig. 3. Fintech exposure measures.
The figure plots the average fintech exposure measures over time. Panel A plots three overall fintech exposure intensities: (i) one constructed using all fintech patent applications ending in a given year (green line), (ii) one constructed using fintech patent applications in the five-year period ending in a given year (red, the default used in our analysis), and (iii) one constructed using granted fintech patents in the five-year period ending in a given year (blue line). Panel B plots the time series of average disaggregated fintech exposure intensities constructed using the subset of fintech patents in each of the seven fintech subcategories in the five-year rolling window. The intensity of fintech exposure is calculated as a standardized mean zero score derived from the raw fintech exposure scores.

(a) Figure 2: fintech patent keywords.

(b) Figure 3: aggregate and disaggregated fintech exposure.

5.3 Figures 4–5: Distinctiveness of fintech exposure and demographics

Read:

- Figure 4 compares fintech exposure with AI, software, routine-task, abstract-task, and offshorability exposure.
- The scatter plots show no simple one-to-one relation between fintech exposure and these broader technology measures.
- Figure 5 shows a nonmonotonic demographic pattern.
- Fintech exposure is high among middle-to-high wage occupations, intermediate-to-high education groups, balanced-gender occupations, and prime-age occupations.

Economic message: Fintech disruption is not the same as routine-task automation, software exposure, or AI exposure. It hits a different part of the labor market: many affected jobs are skilled, finance-adjacent, and mid-career.

Interpretation: This is important for policy. Retraining needs are not limited to low-skill routine workers. Affected workers may already be educated but still need hybrid finance-technology capabilities.

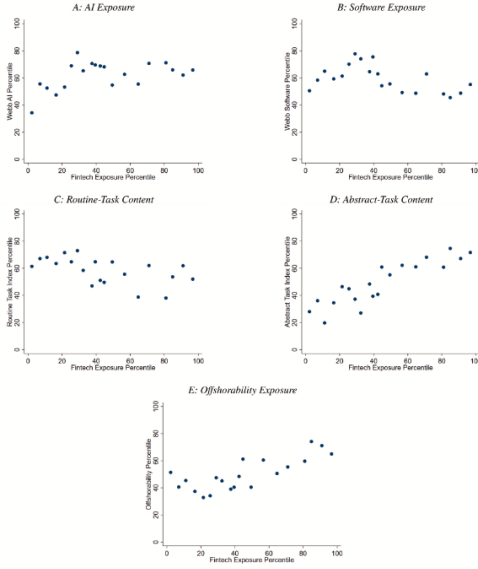


Fig. 4. Fintech exposure vs. General technology exposure. The figure presents the correlation between occupational fintech exposure and five occupational exposure measures including the ones on AI and software constructed by Webb (2019), routine and abstract-task content from Autor and Tyeb (2013), and offshorability from Freije et al. (2011). All data series are at the $\approx 1900d$ occupation level. The authors transform all occupation-level exposure scores to percentile ranks weighted by occupational employment in 2010 and plot the average fintech exposure percentile over the other five exposure measures. The fintech exposure measure is constructed by the authors based on fintech patent filings from 2003 to 2017.

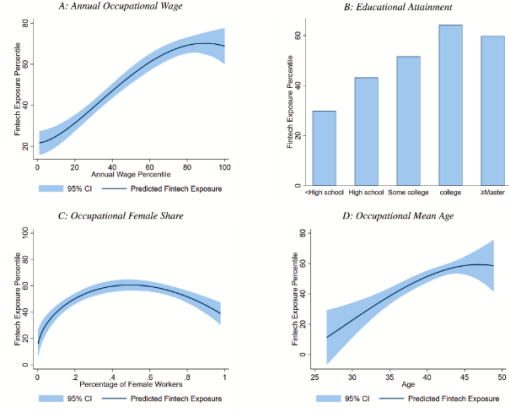


Fig. 5. Fintech exposure by demographic characteristics. The figure plots the average fintech exposure in 2010 over the demographic characteristics of occupations derived from the 2010 American Community Survey (ACS) provided by the Integrated Public Use Microdata Series (IPUMS). Panel A presents the average percentile rank of fintech exposure by the percentile of average annual wages at the occupation level. Panel B presents the average fintech exposure percentile rank across workers' educational attainment. Panel C and D plot the average fintech exposure percentiles by the percentage of females and the average age of workers in each occupation, respectively. All the data series are at the occupation level. Predictions in Panel A, C, and D are estimated using fractional polynomial regressions weighted by occupational employment in 2010. The authors construct the fintech exposure measure based on fintech patent filings from 2003 to 2017. The raw score for fintech exposure and annual wage are transformed into percentile ranks weighted by occupational employment in 2010.

(a) Figure 4: fintech versus other technology exposures.

(b) Figure 5: fintech exposure by demographic characteristics.

5.4 Tables 1–2: Job postings, positions, employment, and wages

Read:

- Table 1 shows a negative relation between fintech exposure and job postings.
- The main controlled panel coefficient is -0.071 , meaning a one-percentile increase in exposure is associated with a 7.1 basis point annual decline in postings.
- Long-difference coefficients are more negative, especially for 2015–2018 and for the full sample period.
- Table 2 validates the result using positions and employment.
- Positions decline by about 6.5 basis points per percentile in the main controlled panel regression.
- Employment also declines, but the magnitude is smaller and less strongly significant. Wages are mostly weak or insignificant.

Economic message: The main disruption shows up first in job postings and positions, then more modestly in employment. This is consistent with firms reducing future demand before realized employment fully adjusts.

Interpretation: The weak wage result is not evidence that fintech is harmless. If workers exit, switch occupations, or if wages are measured only at coarse occupational levels, job quantities can adjust without a clean wage signal.

Table 1
Fintech exposure and job postings.
The table reports the baseline regressions that estimate the effect of fintech exposure on changes in job postings at the occupation ($\alpha \times 1990d$, $\alpha \times \text{year } t$) level. The sample covers job postings based on the BG database from 2010-2018. The dependent variable is 100 times the year-over-year change in the natural logarithm of job posting in columns (1)-(2). The main explanatory variable, fintech exposure measure, is constructed using fintech patent applications in five-year rolling windows. We control for five competing occupational exposure measures to general technology innovations: AI and software exposure developed by Webb (2019), routine and abstract task content by Autor and Dorn (2013) and offshorability from Felps et al. (2011). All the occupation-level exposure scores are transformed to percentile ranks weighted by occupational employment in 2010. We additionally control for year fixed effects and weight the regressions by the underlying postings in the current year in columns (1)-(2). In columns (3)-(5), the dependent variable is the cumulative change in the natural logarithm of job postings over the specified period and the fintech exposure measure is constructed using fintech patent applications published during the corresponding sample period. To improve the precision in columns (3)-(5), we follow the approach proposed by Acemoglu et al. (2022) and use the average of the annual data from 2010-12, 2013-15, and 2016-18 as proxies for postings in 2010, 2015, and 2018, respectively, when calculating the long differences. We also control for the five competing occupational exposure measures to general technology innovations and weight regressions using the averaged underlying postings at the start and end of each period in columns (3)-(5). Standard errors in all columns are clustered at the occupation group level defined by Autor and Dorn (2013). Asterisks denote the significance levels (**+1%, **+=5%, +=10%).

Dep Var	Panel regressions		Long differences		
	$\Delta \log(\text{Postings})_{it} \times 100$		$\Delta \log(\text{Postings})_{it} \times 100$		
	2010-2018		2010-2014	2015-2018	2010-2018
	(1)	(2)	(3)	(4)	(5)
Fintech Exposure _{it-1}	-0.100*** (-3.22)	-0.071*** (-3.46)	-0.125** (-2.08)	-0.153** (-2.26)	-0.330*** (-2.90)
Other Occupational Exposure	No	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	No	No
Observations	2,288	2,288	286	286	286
R ²	0.416	0.433	0.382	0.567	0.670

Table 2
Positions, employment and wages.
The table reports the baseline regressions and estimates the effect of fintech exposure on changes in the number of positions posted, employment, and annual wages at the occupation ($\alpha \times \text{year } t$) level. The sample covers BG posting data in Panel A and the Occupational Employment and Wage Statistics (OEWS) employment and wage data in Panels B and C from 2010-2018, respectively. The dependent variable is 100 times the year-over-year change in the natural logarithm of the number of positions, employment, and annual wages in columns (1)-(2) of Panels A-C, respectively. The main explanatory variable, fintech exposure measure, is constructed using fintech patent applications in five-year rolling windows. All the occupation-level exposure scores are transformed to percentile ranks weighted by occupational employment in 2010. We control for year fixed effects in columns (1)-(2) of each panel and also the five competing occupational exposure measures in the later as in Table 1. We weight the regressions by the underlying positions in the current year in Panel A columns (1)-(2) and by employment in columns (1)-(2) of Panels B and C, respectively. In columns (3)-(5) of each panel, the dependent variable is the cumulative change in the natural logarithm of the outcome variable over the specified period and the fintech exposure measure is constructed using fintech patent applications published during the corresponding period. To enhance precision in columns (3)-(5) of each panel, we follow the approach of Acemoglu et al. (2022) and use the average of the annual data from 2010-12, 2013-15, and 2016-18 as proxies for postings in 2010, 2015, and 2018, respectively, when calculating the long differences. We also control for the five competing occupational exposure measures as in Table 1, and weight regressions using the averaged underlying positions at the start and end of each period in columns (3)-(5) of Panel A and using the averaged underlying employment in columns (3)-(5) of Panels B and C, respectively. We estimate the number of positions in job postings used in Panel A using a textual analysis method outlined in Section A.2 of the Online Appendix. Standard errors are clustered at the occupation group level defined by Autor and Dorn (2013). Asterisks denote the significance levels (**+1%, **+=5%, +=10%).

Panel A: Positions					
Dep Var	Panel Regressions		Long Differences		
	$\Delta \log(\text{Positions})_{it} \times 100$		$\Delta \log(\text{Positions})_{it} \times 100$		
	2010-2018		2010-2014	2015-2018	2010-2018
	(1)	(2)	(3)	(4)	(5)
Fintech Exposure _{it-1}	-0.092*** (-3.38)	-0.065*** (-3.40)	-0.073 (-1.32)	-0.178** (-2.50)	-0.285** (-2.52)
Other Occupational Exposure	No	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	No	No
Observations	2,288	2,288	285	285	285
R ²	0.389	0.402	0.373	0.501	0.617
Panel B: Employment					
Dep Var	Panel Regressions		Long Differences		
	$\Delta \log(\text{Employment})_{it} \times 100$		$\Delta \log(\text{Employment})_{it} \times 100$		
	2010-2018		2010-2014	2015-2018	2010-2018
	(1)	(2)	(3)	(4)	(5)
Fintech Exposure _{it-1}	-0.009 (-1.02)	-0.019* (-1.83)	-0.039 (-1.40)	-0.055* (-2.00)	-0.109* (-1.75)
Other Occupational Exposure	No	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	No	No
Observations	2,288	2,288	286	286	286
R ²	0.017	0.044	0.058	0.044	0.059
Panel C: Annual Wages					
Dep Var	Panel Regressions		Long Differences		
	$\Delta \log(\text{Wages})_{it} \times 100$		$\Delta \log(\text{Wages})_{it} \times 100$		
	2010-2018		2010-2014	2015-2018	2010-2018
	(1)	(2)	(3)	(4)	(5)
Fintech Exposure _{it-1}	-0.003 (-1.50)	-0.001 (-0.47)	0.007 (0.90)	-0.021** (-2.38)	-0.016 (-1.50)
Other Occupational Exposure	No	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	No	No
Observations	2,288	2,288	286	286	286
R ²	0.561	0.573	0.058	0.303	0.109

5.5 Table 3: Complementary versus substitutive fintech exposure

Read:

- Table 3 separates exposed occupations by complementarity.
- The baseline fintech coefficient remains negative for job postings.
- The interaction with above-median complementarity is positive.
- In the quartile specification, higher complementarity quartiles have increasingly positive interactions.
- The pattern is strongest for postings and weaker for employment and wages.

Economic message: Fintech is not purely labor saving. It reduces demand for tasks it substitutes for, but can raise demand for tasks it makes more productive.

Interpretation: This table is the key to interpreting the paper as creative destruction rather than simple destruction. The same technology can destroy routine financial processing tasks while increasing demand for roles that implement, govern, or use fintech systems.

Table 3

Job Complementary vs. Substitutive Effect.

The table reports the regressions that estimate the creative destruction effect of fintech exposure on changes in job postings at the occupation (or year t) level from 2010–2018. The dependent variable is 100 times the year-over-year change in the natural logarithm of the number of positions, employment, and annual wages in Panels A–C, respectively. The main explanatory variable, fintech exposure measure, is constructed using fintech patent applications or a subcategory of fintech patents, in five-year rolling windows. $I(\text{Complementarity} - \text{Above Median})_{it-1}$ equals one for occupations where the employment-weighted percentile rank of their net complementary exposure, calculated as the difference between exposure to complementary fintech patents and substitute fintech patents over the past five years, is above 50 and zero otherwise. We also generate four quartile indicators based on the percentile rank of net complementary exposure, with the bottom (top) quarter representing occupations with the lowest (highest) level of net complementarity. In all regressions, we control for the interactions of the variables, five competing occupational exposure measures and include year dummies as in Table 1. Regressions are weighted using the underlying job postings in the current year in Panel A, and the underlying employment in Panel B and C. All occupational-level exposure scores are transformed to percentile ranks weighted by occupational employment in 2010. Standard errors are clustered at the occupation group level defined by Autor and Dorn (2013). Asterisks denote the significance levels (***)=1%, **=5%, *=10%.

Panel A: Postings				
Dep Var	$\Delta \log(\text{Postings})_{it} \times 100$			
Sample	2010–2018		2015–2018	
	(1)	(2)	(3)	(4)
Fintech Exposure _{it-1}	-0.101*** (-3.89)	-0.176*** (-3.27)	-0.151** (-2.27)	-0.289** (-2.10)
$\times I(\text{Complementarity} - \text{Above Median})_{it-1}$	0.059** (2.41)		0.118* (1.73)	
$\times I(\text{Complementarity} - Q2)_{it-1}$		0.100* (1.76)		0.189 (1.44)
$\times I(\text{Complementarity} - Q3)_{it-1}$		0.137** (2.70)		0.254* (1.84)
$\times I(\text{Complementarity} - Q4)_{it-1}$		0.136** (2.76)		0.276* (1.94)
Other Occupational Exposure	Yes	Yes	Yes	Yes
Year Dummy	Yes	Yes	Yes	Yes
Observations	2,288	2,288	858	858
R ²	0.435	0.437	0.540	0.543
Panel B: Employment				
Dep Var	$\Delta \log(\text{Employment})_{it} \times 100$			
Sample	2010–2018		2015–2018	
	(1)	(2)	(3)	(4)
Fintech Exposure _{it-1}	-0.024* (-2.07)	-0.036*** (-2.86)	-0.022* (-1.93)	-0.022 (-1.20)
$\times I(\text{Complementarity} - \text{Above Median})_{it-1}$	0.007 (0.44)		-0.002 (-0.10)	
$\times I(\text{Complementarity} - Q2)_{it-1}$		0.015 (1.12)		-0.004 (-0.16)
$\times I(\text{Complementarity} - Q3)_{it-1}$		0.019 (0.98)		-0.010 (-0.29)
$\times I(\text{Complementarity} - Q4)_{it-1}$		0.022 (1.20)		0.007 (0.25)
Other Occupational Exposure	Yes	Yes	Yes	Yes
Year Dummy	Yes	Yes	Yes	Yes
Observations	2,288	2,288	858	858
R ²	0.047	0.055	0.045	0.049
Panel C: Annual Wages				
Dep Var	$\Delta \log(\text{Wages})_{it} \times 100$			
Sample	2010–2018		2015–2018	
	(1)	(2)	(3)	(4)
Fintech Exposure _{it-1}	0.002 (0.93)	-0.002 (-0.57)	-0.003 (-0.74)	-0.010 (-1.44)
$\times I(\text{Complementarity} - \text{Above Median})_{it-1}$	-0.004 (-1.55)		-0.004 (-1.16)	
$\times I(\text{Complementarity} - Q2)_{it-1}$		0.007 (1.29)		0.011 (1.64)
$\times I(\text{Complementarity} - Q3)_{it-1}$		-0.001 (-0.13)		0.007 (1.09)
$\times I(\text{Complementarity} - Q4)_{it-1}$		0.002 (0.30)		0.000 (0.04)
Other Occupational Exposure	Yes	Yes	Yes	Yes
Year Dummy	Yes	Yes	Yes	Yes
Observations	2,288	2,288	858	858

Table 3: job complementarity versus substitutive effects.

5.6 Tables 4–5: New occupations and upskilling

Read:

- Table 4 shows that fintech exposure raises the probability that a new suboccupation is created.
- Blockchain has the largest marginal probability among the fintech categories, followed by robo-advising, P2P, data analytics, cybersecurity, and mobile transactions.
- Table 5 shows that fintech exposure increases demand for finance-plus-tech skills and tech-only skills.
- It also increases requirements for more than four years of experience and for a bachelor's degree or higher.
- Finance-only skill demand is not strongly affected.

Economic message: Fintech creates jobs, but not just any jobs. It creates hybrid roles and raises the human-capital threshold for existing occupations.

Interpretation: Table 5 is especially important for workers: surviving the shock requires more than remaining in finance. It requires acquiring technology-related skills that make finance knowledge usable in a fintech environment.

Table 4
Occupation creation amid disruption.

The table reports the Probit regression results based on data at the suboccupation (8-digit SOC code) $\times$ wave level in 2006, 2009, 2011 and 2020. The dependent variable is a dummy that equals one if the suboccupation is newly categorized in a given wave (year) and zero otherwise. The main explanatory variable, fintech exposure measure, is constructed using fintech patent applications or a subcategory of fintech patents, in five-year rolling windows. Fintech exposure of a given suboccupation is measured as the employment-weighted percentile rank of the occupation ($oc_{it}1996d$) to which that suboccupation belongs. In all regressions, we control for the five competing occupational exposure measures as in Table 1. We also include dummies for the occupation group level defined by *Autor and Dorn* (2013), as well as indicators for each wave, except in column (5) where blockchain emerged after 2014. All the occupation-level exposure scores are transformed to percentile ranks weighted by occupational employment in 2010. We report the marginal probabilities corresponding to the coefficient estimates. Standard errors are clustered at the occupation group level. Asterisks denote the significance levels (***=1%, **=5%, *=10%).

Dep Var	[New SOC-8 Occupation] _{year}							
Fintech Subcategory	All	Cyber-security	Mobile Transaction	Data Analytics	Block-chain	P2P	Robo-advising	IoT
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Fintech Exposure _{oc,t-1}	0.011*** (6.46)	0.010*** (4.72)	0.010*** (4.58)	0.011*** (5.79)	0.014*** (3.27)	0.011*** (3.29)	0.011*** (5.86)	0.007*** (3.00)
Marginal Prob	0.119%	0.107%	0.103%	0.115%	0.206%	0.115%	0.120%	0.080%
Other Occupational Exposure	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,968	3,968	3,968	3,968	757	3,968	3,968	3,968
R ²	0.249	0.247	0.246	0.246	0.070	0.244	0.246	0.244

Table 4

(a) Table 4: occupation creation.

Table 5
Upskilling.

The table reports the results of Probit regressions that examine the effect of fintech exposure on skill requirements in job postings at the occupation (o) $\times$ year (t) level from 2010–2018. The dependent variable is an indicator that equals one if a given occupation sees a year-over-year increase in BG job postings that require specific types of skills in columns (1)–(3), four and more year working experiences in column (4), and bachelor and above degree in column (5), and zero otherwise. The main explanatory variable, fintech exposure measure, is constructed using fintech patent applications in five-year rolling windows. In all regressions, we control for the five competing occupational exposure measures and year dummies as in Table 1. All occupation-level exposure scores are transformed to percentile ranks weighted by occupational employment in 2010. Standard errors are clustered at the occupation group level defined by *Autor and Dorn* (2013). We also report the marginal probabilities corresponding to the coefficient estimates in the table. Asterisks denote the significance levels (***=1%, **=5%, *=10%).

Dep Var	[Increases in Postings Requiring] _{o,t}				
	Finance + Tech	Finance only	Tech only	4+ Year Experience	BA+ Education
	(1)	(2)	(3)	(4)	(5)
Fintech Exposure _{oc,t-1}	0.008*** (5.88)	0.001 (0.99)	0.006*** (4.61)	0.003** (2.24)	0.007*** (3.17)
Marginal Prob	0.226%	0.043%	0.150%	0.082%	0.210%
Other Occupational Exposure	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	2,288	2,288	2,288	2,288	2,288
Pseudo R ²	0.131	0.095	0.147	0.096	0.126

(b) Table 5: upskilling.

5.7 Table 6: Sector and fintech-category heterogeneity

Read:

- Panel A shows a negative baseline effect of fintech exposure on postings at the occupation-sector-year level.
- The financial-sector interaction by itself is not always strong, but banking and payments show a positive incremental effect.
- The complementarity interaction is positive, meaning high-complementarity jobs are less harmed or can benefit.
- Panel B shows sharper category-specific effects in the subsectors where each technology is most applicable.
- Cybersecurity, mobile transactions, blockchain, P2P, and robo-advising have large negative baseline effects, but complementarity interactions are positive.

Economic message: Fintech disruptions are largest where the technology is most directly applicable. However, even within affected subsectors, complementary occupations can grow.

Interpretation: This table shows why industry-level averages can be misleading. A bank can simultaneously cut legacy payment-processing jobs and hire workers to build, operate, or supervise new payment technologies.

Table 6

Heterogeneity across Sectors and Fintech Categories.

The table reports the regression results examining the effect of fintech exposure on job postings at the occupation (o) $\times$ sector (NAICS 3-digit, j) $\times$ year (t) level from 2010–2018. The dependent variable is 100 times the year-over-year change in the natural logarithm of job postings. The main explanatory variable, occupational fintech exposure measure, is constructed using all fintech patent applications in five-year rolling windows in Panel A – B, while it is measured based on a subcategory of fintech patents specified in the second row in Panel C. I(Complementarity - Above Median) equals one for occupations where the employment-weighted percentile rank of their net complementary exposure, calculated as the difference between exposure to complementary fintech patents and substitute fintech patents over the past five years, is above 50 and zero otherwise. I(Financial Sector) equals one for NAICS sectors 52 (Finance) and zero otherwise, while I(Tech Sector) equals one for NAICS sectors 51 (Information) and 54 (Professional Services) and zero otherwise following (Acemoglu et al., 2022). We also break the financial sector into three subsectors following (Chen et al., 2019): I(Banking & Payment) equals one for NAICS sectors 522; I(Asset Mgmt. & Brokerage) equals one for NAICS sectors 523; I(Insurance) equals one for NAICS sectors 524. In all regressions, we control for interaction variables by themselves, year fixed effects, and the five competing occupational exposure measures as in Table 1. All the occupation-level exposure scores are transformed to percentile ranks weighted by occupational employment in 2010. Full sample regressions in Panel A additionally control for industry (NAICS 3-digit) fixed effects and are weighted by the industry-year level postings; the subsample regressions in Panel B are weighted by the underlying postings averaged over the prior and current year. Standard errors are double clustered at the occupation group defined by Autor and Dorn (2013) level and industry level in Panel A and at the occupation group level in Panel B. Asterisks denote the significance levels (***)=1%, **=5%, *=10%.

Panel A: By Sector and Job Type						
Dep Var	$\Delta \log(\text{Postings})_{o,j,t} \times 100$					
	(1)	(2)	(3)	(4)	(5)	
Fintech Exposure $_{o,j,t-1}$	-0.053*** (-6.83)	-0.049*** (-4.31)	-0.049*** (-4.19)	-0.069*** (-9.57)	-0.065*** (-6.77)	
$\times$ I(Complementarity - Above Median) $_{o,j,t-1}$				0.051** (2.75)	0.050** (2.63)	
$\times$ I(Financial Sector) $_j$		0.014 (0.56)				
$\times$ I(Banking & Payment) $_j$			0.048* (1.99)		0.046* (1.80)	
$\times$ I(Asset Mgmt. & Brokerage) $_j$			0.026 (1.10)		0.015 (0.57)	
$\times$ I(Insurance) $_j$			-0.015 (-0.86)		-0.018 (-0.64)	
$\times$ I(Tech Sector) $_j$		-0.023 (-1.32)	-0.023 (-1.30)		-0.023 (-1.33)	
Other Occupational Exposure	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	46,238	46,238	46,238	46,238	46,238	
R ²	0.120	0.120	0.120	0.121	0.121	

Panel B: By Fintech Category, Sector and Job type						
Dep Var	$\Delta \log(\text{Postings})_{o,j,t} \times 100$					
Indep Var	Cyber-security	Mobile Transaction	Data Analytics	Blockchain	P2P	Robo-Advising
Subsample	Bank & Payment					Asset Mgmt. & Brokerage
	(1)	(2)	(3)	(4)	(5)	(6)
Fintech Exposure $_{o,j,t-1}$	-0.189** (-2.53)	-0.190** (-2.45)	-0.057 (-1.03)	-0.235** (-2.30)	-0.194*** (-3.22)	-0.343*** (-3.96)
$\times$ I(Complementarity - Above Median)	0.228** (2.34)	0.238** (2.33)	0.080 (1.12)	0.332* (1.76)	0.194** (2.18)	0.245*** (3.88)
Other Occupational Exposure	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	656	656	656	258	656	493
R ²	0.385	0.387	0.378	0.303	0.385	0.329
						0.202

Table 6: heterogeneity across sectors and fintech categories.

5.8 Tables 7–8: Firm-level hiring, separations, and internal transfers

Read:

- Table 7 shows that the negative job-posting effect survives at the firm-occupation-year level.
- Public firms show stronger negative coefficients than the full firm sample.
- Mature and incumbent financial firms appear particularly exposed to incremental declines.
- Table 8 shows that fintech exposure increases internal transfers and separations.
- Transfers often go to occupations with greater complementarity or lower fintech exposure.
- Separations and voluntary separations show similar patterns, suggesting both firm-led and worker-led reallocation.

Economic message: Firms do not simply fire workers in exposed occupations. They also reassign some workers to different roles, especially when workers have finance-plus-tech skills.

Interpretation: Table 8 is a firm-survival mechanism. The firm can adapt by moving labor internally, while a worker without transferable skills may be forced to separate or switch employers.

5.9 Tables 9–10: Innovation responses and firm performance

Read:

- Table 9 shows that firms respond to fintech exposure by increasing fintech innovation through both invention and acquisition.

Table 7

Job postings of entrants, incumbents and mature firms.

The table reports the regressions estimating the effect of fintech exposure on changes in job postings at the firm (i) $\times$ occupation (o) $\times$ year (t) level from 2010 to 2018. The sample covers 67,977 unique firms in the BG database and 3,516 of them are public firms. Columns (1)-(2) are based on all firms, while columns (3)-(8) are based on public firms. The dependent variable is 100 times the year-over-year change in the natural logarithm of job postings. The main explanatory variable, occupational fintech exposure measure, is constructed using fintech patent applications in five-year rolling windows. In columns (2), (4), (6), and (8), we include the interaction of fintech exposure with sector dummies, including the financial sector (NAICS 52) and the tech sector (NAICS 51 and 54), as defined by *Acemoglu et al. (2013)*. In columns (5) and (6), we incorporate the interaction of fintech exposure with an indicator for incumbent firms whose age falls into the top 50%. Additionally, in columns (7) and (8), we include the interaction of fintech exposure with an indicator for mature firms, defined as those whose product life cycle is primarily in the mature or decline stage according to *Hoberg and Malhotra (2012)*. Column (8) features a triple interaction of fintech exposure, sector dummy, and incumbent firm dummy, while column (6) includes a triple interaction of fintech exposure, sector dummy, and mature firm dummy. In all regressions, we control for the five competing occupational exposure measures as in Table 1. All the occupational-level exposure scores are transformed to percentile ranks weighted by occupational employment in 2010. We additionally control for interaction variables by themselves, and firm $\times$ year fixed effects and weight regressions using the firm total postings in the current year. Standard errors are clustered at the occupation group, defined by *Autor and Dean (2013)*, $\times$ firm level. Asterisks denote the significance levels (***)=1%, **=5%, *=10%.

Dep Var	Job Postings (Perf) $\times 100$							
	All Firms		Public Firms					
Sample	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Fintech Exposure _{it-1}	-0.043*** (-6.07)	-0.041*** (-5.57)	-0.063*** (-3.47)	-0.063*** (-3.17)	-0.042*** (-2.94)	-0.048*** (-2.10)	-0.057*** (-4.39)	-0.059*** (-4.45)
$\times$ I(Financial Sector) _{it}		-0.005 (-0.21)		0.001 (0.03)		0.130*** (3.45)		0.146*** (2.32)
$\times$ I(Financial Sector) _{it} $\times$ I(Incumbent) _{it}						-0.144*** (-2.93)		
$\times$ I(Financial Sector) _{it} $\times$ I(Mature Firm) _{it}								-0.145** (-2.01)
$\times$ I(Tech Sector) _{it}		-0.016 (-0.88)		-0.003 (-0.13)		-0.006 (-0.21)		0.001 (0.04)
$\times$ I(Incumbent) _{it}					-0.031* (-1.80)	-0.023 (-1.31)		-0.018 (-0.74)
$\times$ I(Mature Firm) _{it}							-0.019 (-0.96)	-0.018 (-0.74)
Other Occupational Exposure	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,002,968	1,002,968	213,817	213,817	213,413	213,413	213,817	213,817
Pseudo R ²	0.76	0.76	0.761	0.761	0.764	0.764	0.761	0.761

- The relation is especially pronounced for financial firms; technology firms are more closely linked to invention than acquisition in some specifications.
- Growth firms are more likely to invent fintech patents when exposed.
- Table 10 shows that fintech exposure alone does not guarantee better outcomes.
- Public firms with fintech inventor status show stronger employment, sales, and TFP performance.
- Firms whose workforces are largely complementary to fintech also show better employment, sales, and productivity outcomes.

Economic message: The firm-level winners are not all exposed firms. The winners are exposed firms that either innovate or have a workforce whose tasks complement fintech.

Interpretation: These tables complete the paper's mechanism: disruption first changes labor demand, then firms respond through innovation and reallocation, and only firms with the right capabilities turn the shock into operating gains.

Table 9

Firm innovation responses.

The table reports Poisson regressions examining the effect of fintech exposure on innovation activities at the firm (i) $\times$ year (t) level. The sample includes both public and private firms in columns (1)-(4) and only public firms in columns (5)-(8) covered by the Compustat and BG data in 2010-2017, respectively. The dependent variable is the number of fintech patents invented in columns (1)-(2) and the number of fintech patents acquired in columns (3)-(4) and (7)-(8), respectively, identified from the USPTO data. The main explanatory variable, lagged fintech exposure of a given firm, is measured using the average annual fintech exposure of its industry (NAICS 3-digit) over a three-year rolling period. The annual industry-level fintech exposure is calculated as the employment-weighted average of occupational fintech exposure percentiles constructed using fintech patent filings in the five-year rolling windows. The employment data at the industry-occupation level is from the ONS database. In columns (2), (4), (6), and (7), we interact the lagged industry-level fintech exposure with sector dummies including the financial sector (NAICS sector 52) and the tech sector (NAICS sectors 51 and 54). In columns (6) and (8), we interact the lagged industry-level fintech exposure with an indicator for growth firms whose lagged Tobin's Q falls in the top quartile among all public firms. Firm attributes include interaction variables by themselves and lagged employment of public firms at job postings of private firms. We additionally control for interaction variables by themselves, the industry (NAICS 3-digit) by year fixed effects. Standard errors are clustered at the industry level. We also report the marginal effects corresponding to the coefficient estimates in the table. Asterisks denote the significance levels (***)=1%, **=5%, *=10%.

Dep Var	All Firms				Public Firms			
	Number of Fintech Patents _{it}							
Sample	Invented		Acquired		Invented		Acquired	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Fintech Exposure _{it-1}	0.187*** (5.48)	0.109*** (2.24)	0.128*** (6.33)	0.113*** (3.42)	0.105*** (6.40)	0.179*** (4.84)	0.142*** (6.09)	0.130*** (3.70)
$\times$ I(Financial Sector) _{it}		0.218*** (14.42)		0.103*** (2.09)		0.258*** (14.40)		0.124*** (6.03)
$\times$ I(Tech Sector) _{it}		0.069*** (4.81)		-0.005 (-0.15)		0.077*** (4.99)		-0.044* (-1.92)
$\times$ I(Growth Firm) _{it-1}						0.106** (2.55)		0.023 (0.36)
Incremental effect on DV								
Fintech Exposure	0.22%	0.13%	0.04%	0.04%	1.06%	1.79%	0.33%	0.51%
$\times$ I(Financial Sector)		0.25%		0.04%		0.29%		-0.10%
$\times$ I(Tech Sector)		0.08%		0.00%		0.78%		0.09%
$\times$ I(Growth Firm)					1.12%			
Firm Attributes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	194,483	194,483	116,843	116,843	21,844	20,314	26,094	19,314
Pseudo R ²	0.415	0.419	0.285	0.287	0.270	0.348	0.196	0.214

(a) Table 9: innovation responses.

Table 8

Separation and within-firm transfers.

The table reports Poisson regressions examining the effect of fintech exposure on job transfers within firms and separations. The sample covers employees of public firms in the financial and tech sectors from 2010 to 2018, based on LinkedIn data. The analysis is performed at the occupation (o) $\times$ firm (i) $\times$ year (t) level. In Panel A, the dependent variable is the number of internal transfers in different occupations (employees move to a different occupation within the same firm) in a given year. In Panel B, the dependent variable is the number of job separations to different occupation (employees leave for another firm and a different occupation) in a given year. In Panel C, the dependent variable is the number of voluntary separations in five-year rolling windows, where the destination firm has no fewer employees than the originating firm or the destination position's seniority level is the same or higher than that of the originating position based on Revella's seniority metric. In all three panels, the first two columns analyze job moves to all kinds of destination occupations, while columns (3) and (4) focus on moves to occupations with higher fintech complementarity percentile ranks, and columns (5) and (6) focus on moves to occupations with lower fintech exposure percentile ranks. The main explanatory variable is the lagged fintech exposure measure of the originating occupation constructed using fintech patent applications in five-year rolling windows. I% Fin + Tech Skill - Q4) is equal to one if the proportion of employees with both finance-related and tech-related skills within an occupation-firm cohort falls in the top quartile for a given year. In all regressions, we control for the five competing occupational exposure measures and year dummies as in Table 1. All the occupation-level exposure scores are transformed to percentile ranks weighted by occupational employment in 2010. We additionally control for interaction variables by themselves, the lagged employment of a given firm-occupation-year cohort, the occupation group fixed effects, and firm $\times$ year fixed effects. Standard errors are clustered at the occupation group $\times$ firm level. Asterisks denote the significance levels (***)=1%, **=5%, *=10%.

Dep Var	Panel A: Internal Transfers _{it}					
	All		To Greater Complementarity		To Lower FT Exposure	
	(1)	(2)	(3)	(4)	(5)	(6)
Fintech Exposure _{it-1}	0.011*** (13.63)	0.007*** (8.22)	0.010*** (9.45)	0.007*** (6.13)	0.050*** (31.29)	0.043*** (29.63)
$\times$ I% Fin + Tech Skill - Q4) _{it-1}				0.006*** (3.84)		0.023*** (4.85)
Incremental effect on DV						
Fintech Exposure	2.59%	1.59%	1.22%	0.79%	5.16%	4.49%
$\times$ I% Fin + Tech Skill - Q4)		1.85%		0.70%		2.45%
Occupational Controls	Yes	Yes	Yes	Yes	Yes	Yes
Occupation Group FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	187,546	187,546	155,474	155,474	155,168	155,168
Pseudo R ²	0.474	0.479	0.37	0.373	0.500	0.504

Dep Var	Panel B: Separations _{it}					
	All		To greater complementarity		To lower FT exposure	
	(1)	(2)	(3)	(4)	(5)	(6)
Fintech Exposure _{it-1}	0.015*** (14.98)	0.009*** (7.90)	0.014*** (11.95)	0.009*** (7.90)	0.057*** (32.08)	0.048*** (30.40)
$\times$ I% Fin + Tech Skill - Q4) _{it-1}				0.008*** (4.67)		0.027*** (5.84)
Marginal Incremental effect on DV						
Fintech Exposure	1.17%	0.71%	0.60%	0.39%	2.18%	1.84%
$\times$ I% Fin + Tech Skill - Q4)		0.65%		0.29%		1.04%
Occupational Controls	Yes	Yes	Yes	Yes	Yes	Yes
Occupation Group FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	174,649	174,649	145,826	145,826	144,207	144,207
Pseudo R ²	0.403	0.41	0.304	0.308	0.441	0.445

Dep Var	Panel C: Voluntary Separations _{it}					
	All		To greater complementarity		To lower FT exposure	
	(1)	(2)	(3)	(4)	(5)	(6)
Fintech Exposure _{it-1}	0.015*** (15.59)	0.009*** (9.57)	0.014*** (12.57)	0.009*** (7.92)	0.057*** (32.75)	0.048*** (29.94)
$\times$ I% Fin + Tech Skill - Q4) _{it-1}				0.007*** (3.70)		0.029*** (6.36)
Marginal Incremental effect on DV						
Fintech Exposure	1.02%	0.60%	0.54%	0.35%	1.91%	1.60%
$\times$ I% Fin + Tech Skill - Q4)		0.61%		0.26%		0.97%
Occupational Controls	Yes	Yes	Yes	Yes	Yes	Yes
Occupation Group FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes

6 Short summary

- The paper studies fintech as a recent wave of creative destruction in finance-related labor markets.
- It builds a novel fintech exposure measure by matching O*NET task text to fintech patent text.
- Fintech exposure is distinct from AI, software, routine-task, abstract-task, and offshorability exposure.
- High-exposure occupations experience declines in job postings and employment.
- Wages show weaker effects, suggesting quantities and job definitions adjust more clearly than occupational wages.
- The negative effect is stronger in later years and in categories where fintech patents are most applicable.
- Complementary occupations experience weaker declines or gains.
- Fintech exposure increases the creation of new suboccupations.
- Fintech increases demand for finance-plus-tech skills, tech skills, experience, and education.
- Firms adapt through changed hiring, internal transfers, separations, and innovation.
- Financial incumbents and mature firms are especially vulnerable if their legacy jobs are substitutable.
- Innovative firms and firms with fintech-complementary workforces can grow employment, sales, and productivity.

7 Conclusion

7.1 Core conclusion

Jiang, Tang, Xiao, and Yao show that fintech is both a labor-displacing and opportunity-creating technology shock. At the occupation level, exposure to fintech patents is associated with lower job postings and employment. At the same time, fintech exposure is associated with new suboccupation creation, higher skill requirements, internal labor reallocation, and firm-level innovation. The central message is therefore not that fintech simply eliminates jobs, but that it changes which tasks, skills, and firms are valuable.

7.2 Worker-level interpretation

The paper’s results are less comforting for individual workers than for firms. Occupations exposed to fintech face net declines in postings and employment, and the most affected workers are not necessarily the least educated or lowest paid. Fintech often targets middle-skill or skilled financial tasks: payments processing, verification, data handling, financial analysis, compliance, credit evaluation, and customer-facing financial intermediation. Workers with purely traditional finance skills can therefore be vulnerable even if they have substantial education or work experience.

The upskilling results imply that worker adjustment depends on skill bundling. Finance knowledge remains useful when combined with software, data, cybersecurity, or platform skills. But finance skills by themselves do not show the same increase in demand. This is why the paper’s labor-market implication is not merely “learn to code.” It is that technology-specific finance workers become more valuable when they can translate domain knowledge into fintech implementation, governance, and adoption.

7.3 Firm-level interpretation

Firms are more flexible than occupations. A firm can reduce hiring in obsolete jobs, shift workers internally, create new roles, acquire fintech capabilities, or develop its own fintech patents. This is why the negative occupation-level effect does not mechanically imply negative firm-level performance. Firms with innovative capacity or complementary workforces can turn fintech exposure into growth in employment, sales, and productivity.

The firm results also clarify why incumbents do not all lose. Mature financial firms can be harmed when legacy jobs are directly substituted, but incumbents with innovation capacity may survive by converting technological threat into organizational change. The relevant distinction is not simply old versus new firms; it is whether the firm’s workforce and innovation strategy are complementary to the new technology.

7.4 Interpretation for creative destruction

The paper provides a nuanced version of creative destruction. Destruction appears in the decline of exposed job postings and employment. Creation appears in new suboccupations, hybrid skill requirements, fintech patents, and productivity gains for adaptive firms. The same shock can therefore be destructive for one worker, neutral for another, and positive for a firm that reallocates resources.

This also helps reconcile conflicting narratives about fintech. One narrative says fintech automates financial intermediation and reduces jobs. Another says fintech expands access, creates platforms, and increases demand for digital finance talent. The paper shows that both narratives are correct, but they apply to different task types and firms.

7.5 Policy and training implications

The policy implication is that training programs should focus on complementarity rather than generic reskilling. The goal is not only to help workers leave finance, but to help finance workers acquire technology skills that make their existing domain knowledge productive. Practical training categories include data analytics, cybersecurity, digital payments, blockchain infrastructure, API-based financial services, product management, compliance technology, and risk controls for automated systems.

The results also suggest that labor-market policies should anticipate within-firm mobility. If firms can redeploy workers internally, then training incentives and worker-matching systems can be designed around internal transitions, not only unemployment spells. Internal transfer pathways may be especially useful for workers whose existing occupations are exposed but whose firm is expanding in complementary fintech roles.

7.6 Managerial implications

For managers, the paper implies that fintech exposure should be treated as a workforce portfolio problem. Firms need to identify which occupations are substitutes, which are complements, and which jobs require skill-bundling. A fintech strategy that only purchases technology may fail if the workforce cannot absorb it. Conversely, a firm with finance-plus-tech talent can use disruption as a productivity-enhancing opportunity.

The paper also shows why innovation strategy matters. Firms that invent or effectively acquire fintech capabilities perform better than firms that merely face the shock. This implies that managers should integrate human-capital planning with innovation and acquisition strategy.

7.7 Limitations and open questions

Several open questions remain. The paper measures jobs and firm outcomes, but the welfare consequences for displaced workers require more evidence on earnings losses, retraining success, and long-run career paths. The patent-based measure is powerful but may understate non-patented platform innovations or open-source technologies. The study also leaves room for more granular evidence on which specific training programs convert substitutable finance workers into complementary finance-plus-tech workers.

A further question is distributional. If fintech creates high-skill hybrid jobs while eliminating middle-skill processing jobs, the gains may accrue to firms and already-skilled workers, while adjustment costs fall on workers whose skills are close to but not complementary with fintech. Future research could study who captures the surplus from fintech adoption: workers, firms, consumers, or technology vendors.

7.8 Final takeaway

The paper's final takeaway is that the survival margin is complementarity. Workers survive fintech disruption when their skills complement the new technology. Firms survive and grow when they can reallocate labor and innovate around fintech. Fintech therefore does not simply replace finance labor; it reorganizes finance work around data, software, platforms, and hybrid human capital.